

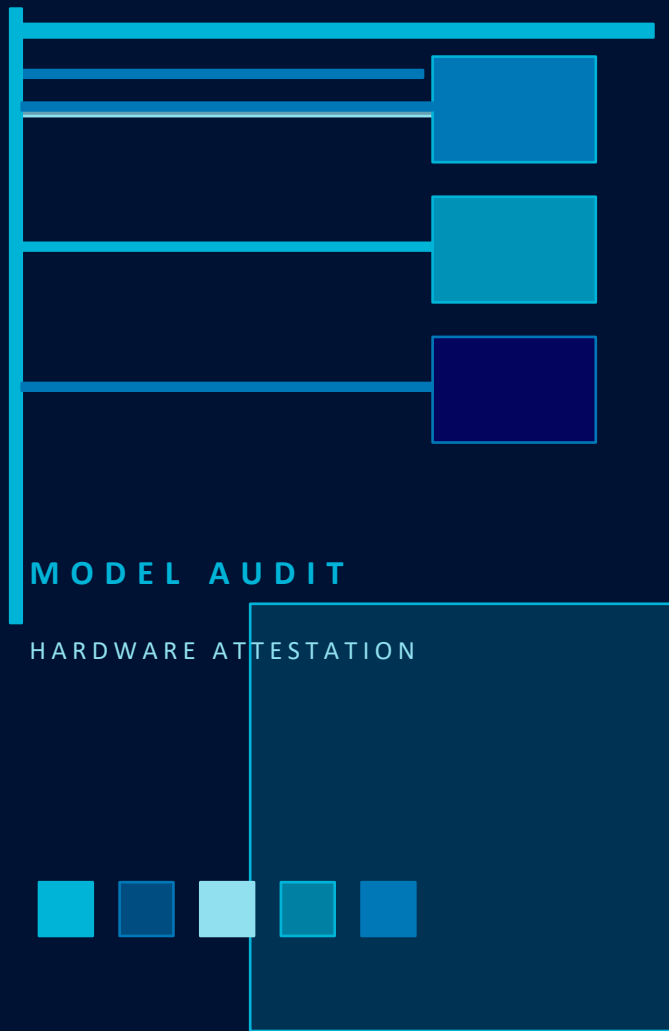
Are You Getting What You Pay For?

Auditing Model Substitution in LLM

APIs

Why Software Tests Fail and Hardware Attestation Wins

arxiv.org/abs/2504.04715



The Model Substitution Problem: Economic Incentives Drive Covert Swaps

THE PROBLEM

Commercial LLM API providers charge premium rates for specific models, but the black-box nature of inference endpoints provides no cryptographic guarantee that the billed model is the one actually running.

Strong economic incentives exist for covert substitution: deploying cheaper quantized or smaller models can cut GPU costs by 4–10x while appearing functionally similar to the specified model in casual use.

Formalized as hypothesis testing:

H_0 — Specified model is being served

H_1 — An alternative model has been substituted

1

Smaller Models

8B served instead of 70B —
significant compute savings at
quality cost

2

Quantized Variants

FP8 / INT8 weights instead of FP16
— lower GPU memory footprint

3

Fine-tuned / Updated

Silently swapped fine-tunes or
checkpoint updates after contract

4

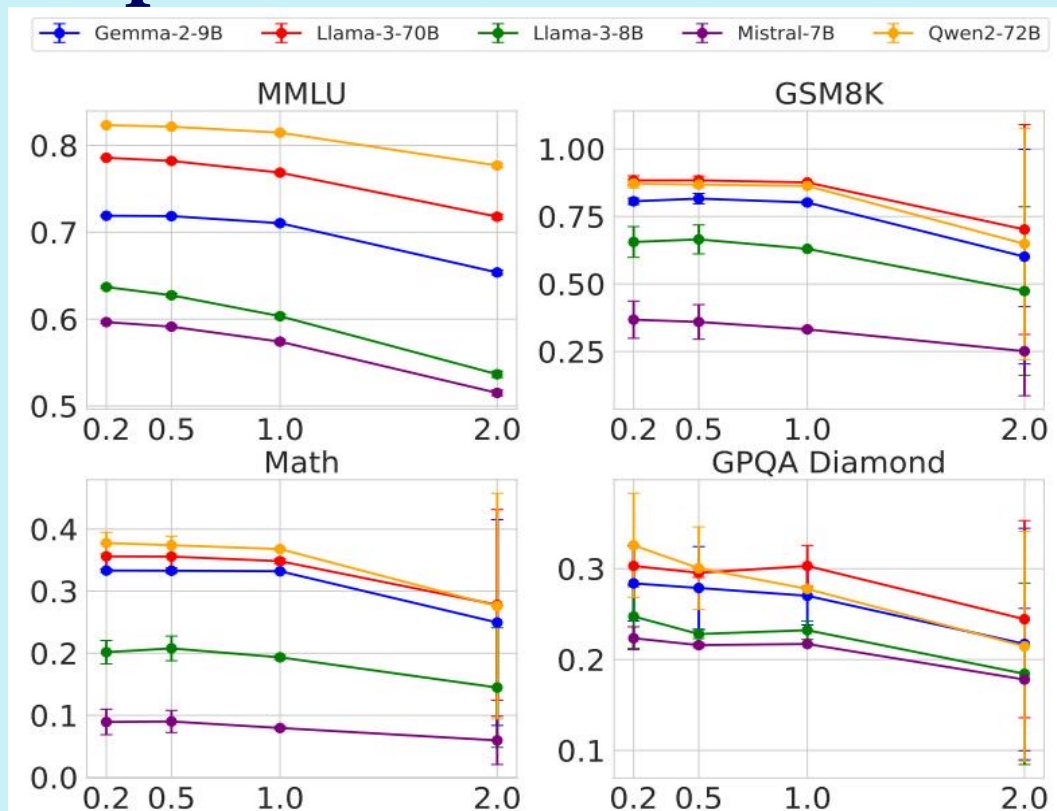
Different Model Family

Entirely unrelated architecture
substituted for the billed model

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Higher Temperature Amplifies Performance Gaps but Increases Noise



Temperature Scaling Effect

At temperature 2.0, performance degrades across all models and benchmarks.

Detection Challenge

Higher temps degrade both original and FP8 models similarly, narrowing the detectable gap.

Key Insight

Temperature variation adds noise that overwhelms the already-small quantization signal.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Detection Method

Compare token-level log probabilities between local reference model and API response.

The Nondeterminism Problem

Same Llama-3-8B model produces varying log-probs across:

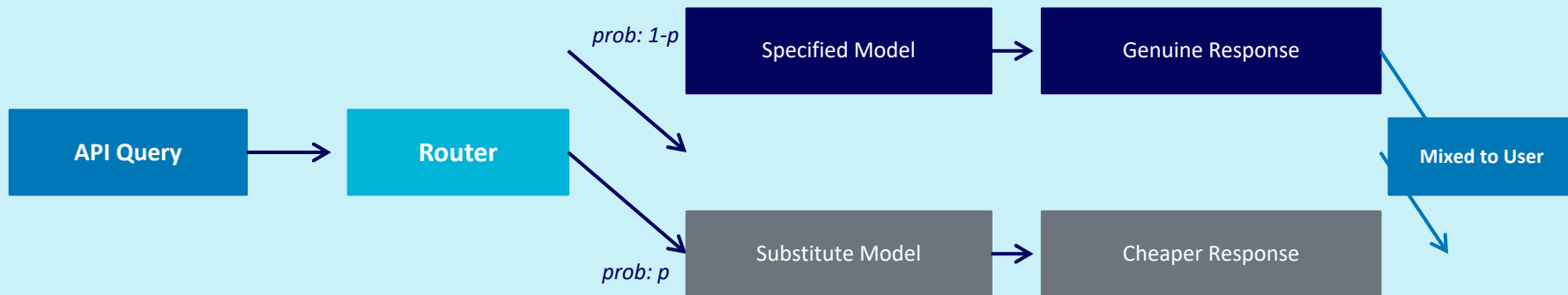
- transformers 4.40 vs 4.50
- vLLM 0.6.2 vs 0.8.2
- A100 vs H100 GPU hardware

Consequence

Auditor cannot distinguish legitimate model on different infra from actual substitution.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive



Statistical Undetectability

As $p \rightarrow 0$, mixed distribution converges to P_{spec} . Tests need exponentially more queries.

Benchmark Evasion

Detect benchmark prompts, route to genuine model. Software audits see only genuine responses.

Software-Only Auditing Methods Are Fundamentally Unreliable

ALL THREE METHODS FAIL UNDER REALISTIC CONDITIONS

Method Type	Core Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Requires large query budgets; fails against subtle substitutions	At substitution rate $p \rightarrow 0$, tests require exponentially more samples to reach significance
Log-Probability Fingerprinting	Production nondeterminism across software versions and GPU hardware	Cannot distinguish legitimate model on different infra from actual substitution
Adversarial Countermeasures (Benchmark Evasion / Randomized Routing)	Provider detects and routes around auditing attempts	Any software-based approach can be gamed once its detection logic is known

Source: arxiv.org/abs/2504.04715

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

PROVABLE

Cryptographic attestation guarantees correct weights & code

EFFICIENT

Modest performance overhead vs ZKPs which are computationally prohibitive

ROBUST

Hardware-level security cannot be evaded by routing or software tricks

Software Methods ✗

- Defeated by inference nondeterminism
- Query-prohibitive for subtle substitutions
- Gameable via adversarial routing

TEE Hardware Attestation ✓

- Cryptographically verifiable model identity
- Practical: NVIDIA H100 Confidential Computing
- No software audit needed

KEY TAKEAWAYS

Close the Verification Gap with Hardware Attestation

1

Model substitution is economically motivated and covert — providers have strong incentives to replace advertised models with cheaper alternatives.

2

FP8 quantization barely affects benchmark scores — making it a nearly undetectable substitution method.

3

Inference nondeterminism defeats log-probability fingerprinting — same model produces different log-probs across software versions and GPU hardware.

4

Randomized substitution defeats statistical tests — as substitution rate $p \rightarrow 0$, detection requires exponentially more queries.

5

Trusted Execution Environments are the only currently viable path — TEEs provide cryptographic, hardware-verified model integrity with modest overhead.

The community must move beyond software-only auditing and adopt hardware attestation standards for verifiable LLM APIs.